

Battery Day-Ahead Arbitrage

STRUCTURAL

Standalone storage optimised against the daily auction curve

Jan 2019 – Jun 2026

Source: EPEX SDAC day-ahead auctions via Energy-Charts (Fraunhofer ISE, CC BY 4.0). Backtest by author.

2025 REV / MW

€85.9k

DE-LU

2025 REV / MW

€88.4k

CZ

2019–25 AVG / MW

€63.1k

DE-LU

2026 YTD ANN.

€89.9k

DE-LU run-rate

PEAK YR (2022)

€122.7k

gas crisis

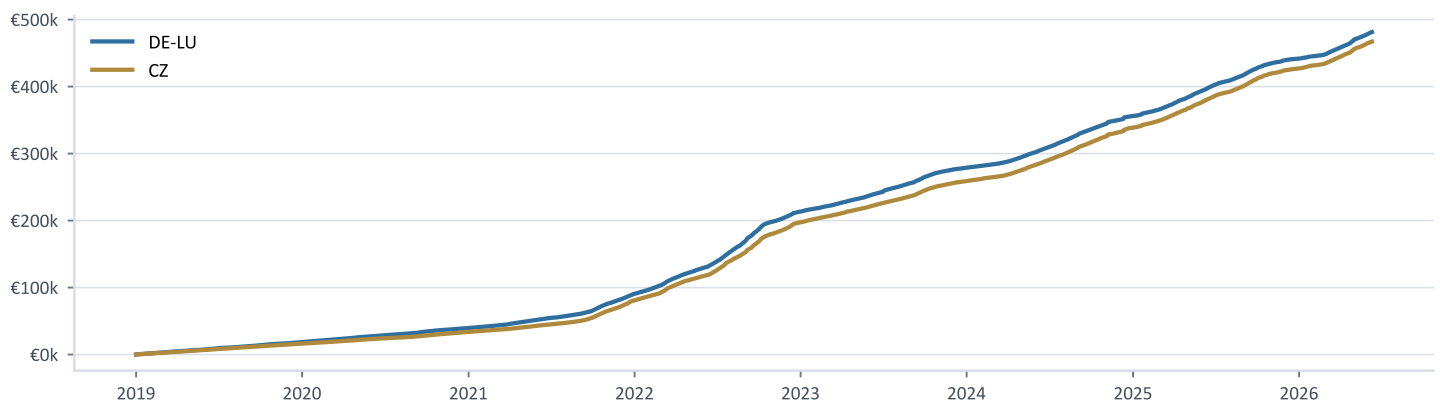
STRATEGY & EDGE

A 1 MW / 2 MWh battery buys the cheapest hours and sells the dearest hours of each day's EPEX day-ahead auction. The schedule is the exact profit-maximising dispatch, solved as a linear program (HiGHS) subject to power, capacity, round-trip efficiency and cycle-life limits, net of exchange fees.

The edge is structural rather than statistical: every additional GW of solar deepens the midday price trough while the morning and evening ramps hold firm, so the intra-day spread a battery harvests widens year after year. The natural counterparties are must-run solar selling at any price midday and inflexible demand buying the evening peak.

Cumulative arbitrage revenue per MW installed

EUR per MW, day-ahead only, costs included



HOW IT WORKS

- Inputs: the 96 quarter-hourly (historically 24 hourly) cleared prices for the delivery day.
- Optimise charge/discharge per period; state of charge bounded by usable energy, throughput bounded to ≤ 2 cycles/day.
- Revenue booked at cleared prices minus 0.10 €/MWh fees on every traded MWh.
- Day-ahead figure is the floor: operators stack intra-day and ancillary services on top, typically +30–80%.
- Perfect within-day foresight; a live bidder gives up a few percent to forecast error against the cleared curve.
- CZ now out-earns DE-LU — rising Czech solar, minimal domestic storage to compete it away.

ASSUMPTIONS & SPECIFICATION

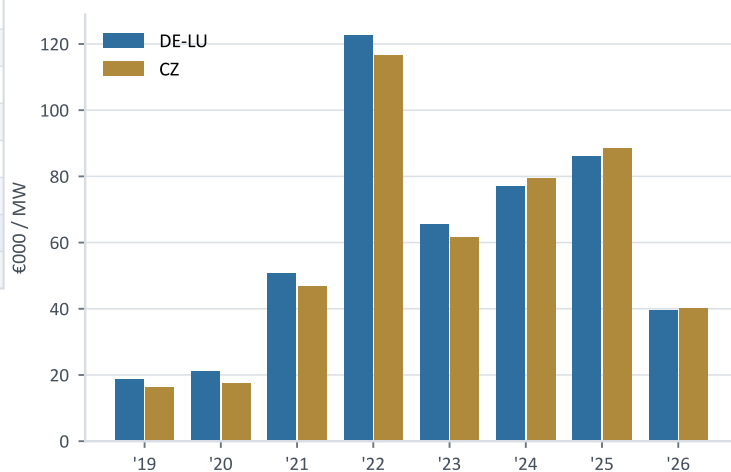
1 MW / 2 MWh (2-hour) lithium system · 86% round-trip efficiency (0.927 each way) · ≤ 2.0 full cycles per day enforced as a throughput cap · 0.10 €/MWh exchange + clearing fee per MWh traded · no degradation/financing modelled · revenue is gross of capex and fixed O&M, expressed per MW of installed power so it scales linearly.

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REVENUE PER MW BY YEAR

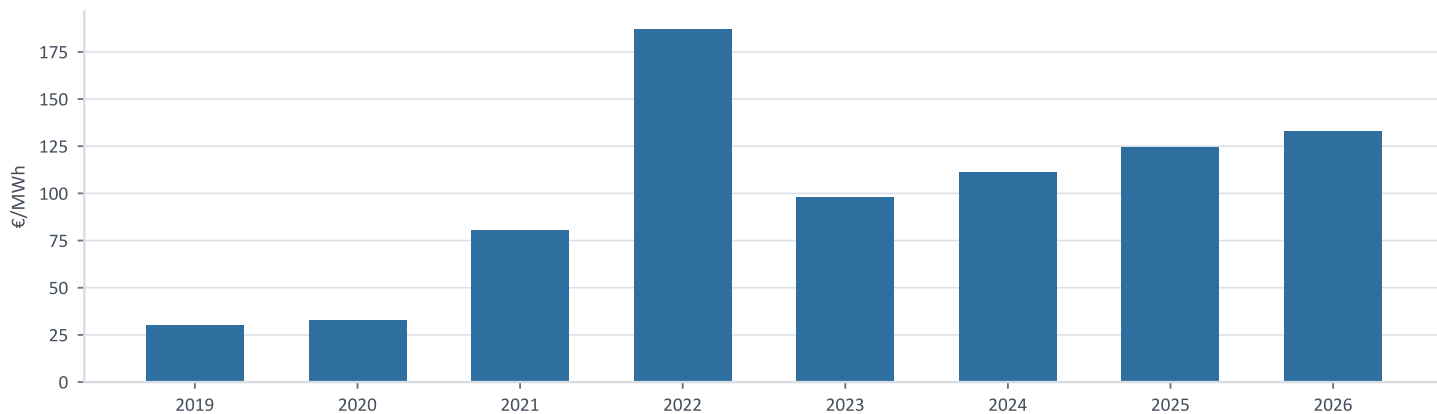
Year	DE-LU	CZ	Avg spread	Days
2019	€18.6k	€16.4k	30	365
2020	€21.1k	€17.4k	32	366
2021	€50.8k	€46.9k	80	365
2022	€122.7k	€116.6k	187	365
2023	€65.5k	€61.7k	98	365
2024	€77.0k	€79.4k	111	366
2025	€85.9k	€88.4k	124	365
2026 YTD	€39.7k	€40.2k	133	161

Annual revenue per MW



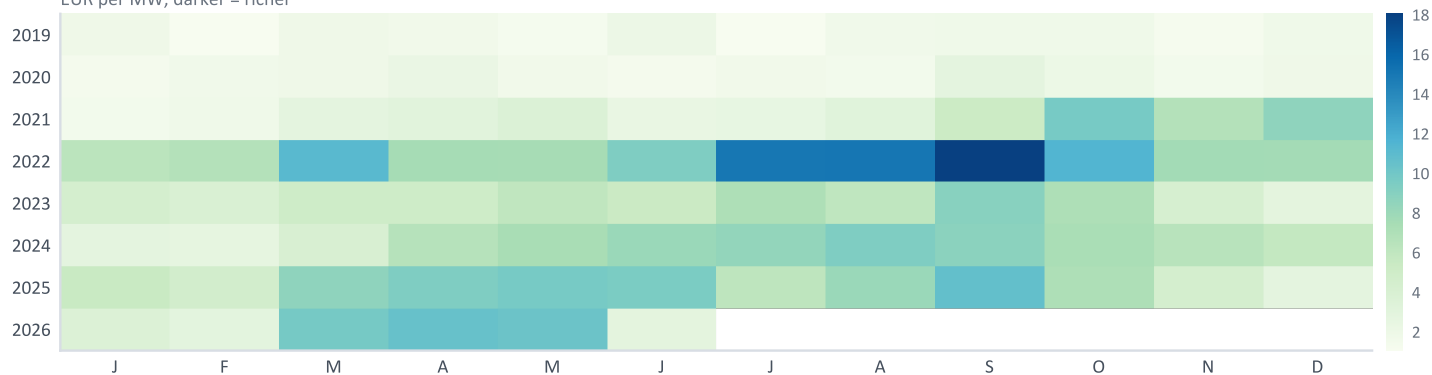
The driver: average daily price spread (DE-LU)

Mean daily max-min of the day-ahead curve, EUR/MWh



Monthly revenue seasonality (DE-LU)

EUR per MW, darker = richer



KEY RISKS & EDGE DECAY

Edge decay is the central risk: the spread is harvested by every new battery, so revenue per MW falls as the storage fleet catches up with solar — watch German/Czech storage build-out and the annual revenue bars as the monitor. · Revenue is volatile: 2022's gas crisis tripled spreads; a return to calm fundamentals compresses them. · Realisation risk: live bids precede the clear, so forecast error shaves a few percent off the LP optimum. · Not modelled: degradation, augmentation capex, grid fees, balancing exposure.